Locating the epicentre of an earthquake 9

Shown here is a map locating Shaketown and some of the surrounding towns that have seismograph stations. Seismographs are used to record the strengths of earthquakes. Information has been sent to Shaketown about the recent earthquake from three seismic stations. All information has been pooled and recorded in Table 1. You have been asked to analyse the information and work out where the epicentre of the earthquake is by drawing it on the map. The epicentre is directly above the origin of the earthquake. Work through the instructions below and locate the epicentre.

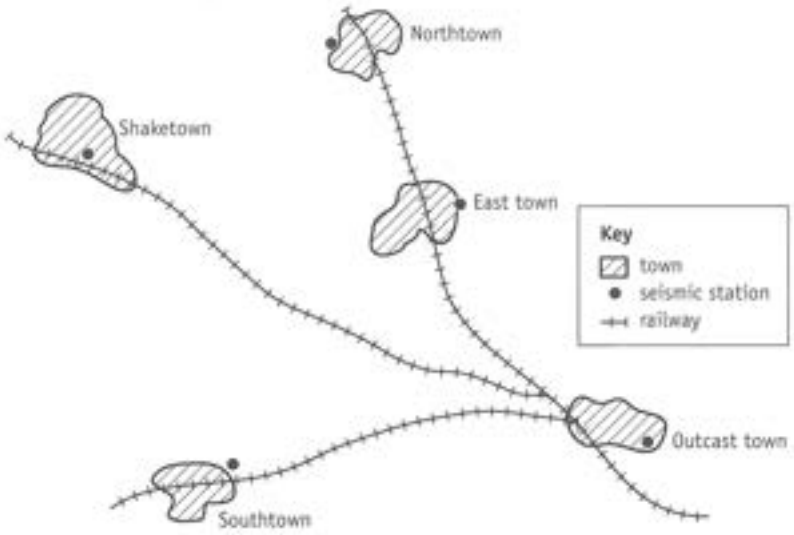


Table 1

Location of station	Arrival time of P waves	Arrival time of S waves	Time difference	Distance from epicentre	Radius of circle
Shaketown	07.06.10	07.07.30	1 min 20 s	1500 km	1.5 cm
Northtown	07.09.15	07.11.55			
East town	07.10.05	07.14.45			
Southtown	07.15.19	07.20.41			

1 Calculate the difference between the time of arrival of the P and S waves. The first one has been done for you. Write your answers in the 'Time difference' column of Table 1.

2 Use Table 2 to calculate the distance from the epicentre. Add your answers in the 'Distance from epicentre' column of Table 1.

Table 2

Difference in times between P and S waves	Distance in kilometres
40 seconds	500
1 minute 20 seconds	1500
2 minutes 40 seconds	2500
3 minutes and 20 seconds	3500
4 minutes and 40 seconds	4500
5 minutes and 20 seconds	5500

3 Calculate the radius of the circle using the scale 1 cm = 1000 km. Add your answers in the 'Radius of circle' column in Table 1.

4 On the map above, draw circles using a compass to indicate the distances. The seismic station will be the centre of the circle and the distance will be the radius measurement from Table 1. Draw the circles around each of the seismograph stations.

5 The epicentre is where all the circles meet. Mark the epicentre with a red cross on the map.

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